

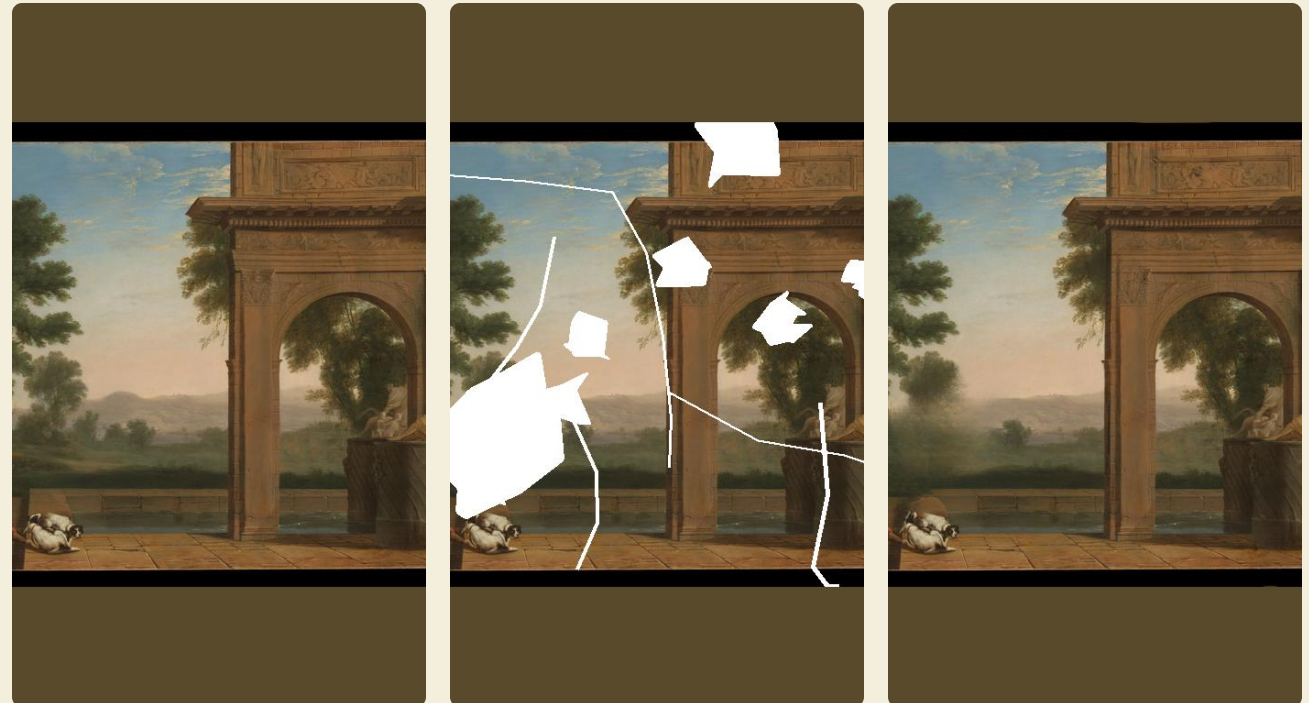
Motivation

A plausible image can still be a poor restoration.

Restoration quality is local, multi-dimensional and easy to misread.

Traditional whole-image similarity can reward a visually smooth result while hiding errors inside the repaired area or along its boundary.

- **Where did the image change?**
damaged region, boundary and unchanged context
- **What kind of evidence changed?**
pixels, perception, features, texture, colour and seams
- **Can the result be trusted?**
inspect stability and failure signals before acceptance



REFERENCE

Clean digital reference

CONTROLLED DAMAGE

Mixed missing-region damage

RESTORATION CANDIDATE

LaMa candidate for inspection

CORE POSITION

The framework supports transparent inspection of digital candidates. It does not approve physical conservation treatment.

Research Questions

Three questions connect measurement, comparison and stability.

RQ1

Evidence beyond similarity

What additional evidence does a region-aware, multi-metric evaluation framework provide beyond traditional image-similarity metrics when evaluating AI-assisted painting restoration?

REGIONS +
METRICS

RQ2

Method differences

How do selected inpainting methods differ in restoration quality across controlled artificial damage conditions, and how consistent are these differences across the evaluated paintings?

PAIRED
COMPARISON

RQ3

Stochastic stability

How can repeated-candidate disagreement be used to characterize the stability of stochastic painting restorations, and how does it relate to other restoration-quality diagnostics?

SEEDS +
DIAGNOSTICS

The questions deliberately separate candidate quality from candidate consistency.

Methods

*Controlled inputs, complementary evidence,
traceable decisions.*



1 PREPARE

Normalize to 768×768 RGB
Preserve aspect ratio and content
bounds



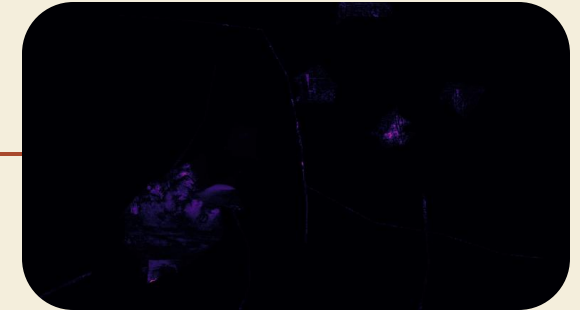
2 DAMAGE

Canonical masks
Size and geometry variants
Procedural degradations



3 RESTORE

Telea · LaMa · HINT
Stable Diffusion 1.5
Bounded SDXL branch



4 DIAGNOSE

Absolute error vs original
Bright pixels = mismatch
Dark pixels = closer match

EVIDENCE REMAINS SEPARATE

REFERENCE

pixel error · PSNR · SSIM

PERCEPTUAL

LPIPS · CLIP · DINOv2

LOCAL

texture · direction · colour · seam

SPATIAL

error · improvement · uncertainty
maps

ANALYSIS DISCIPLINE

Paintings are the independent unit. Shared cases use paired comparisons; seeds, masks and cases are treated as repeated or nested evidence. No universal quality, uncertainty or trustworthiness score is constructed.

Data

A balanced controlled collection, designed to scale.

300

target paintings

60

per visual category

768 × 768

RGB analysis canvas

GROUPING BOUNDARY

Five broad visual categories support balanced comparison. They are not treated as validated art-historical styles.



PORTRAIT / FIGURE

p001 Juan de Pareja



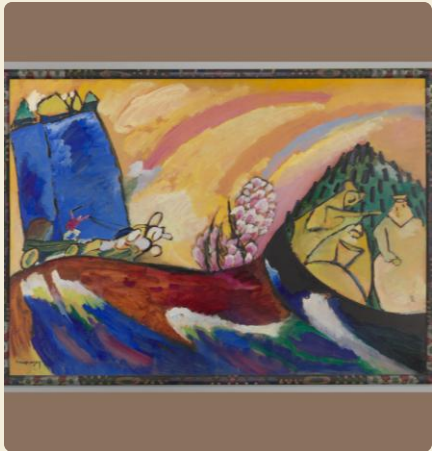
LANDSCAPE / NATURAL

p018 Classical Landscape with Figures



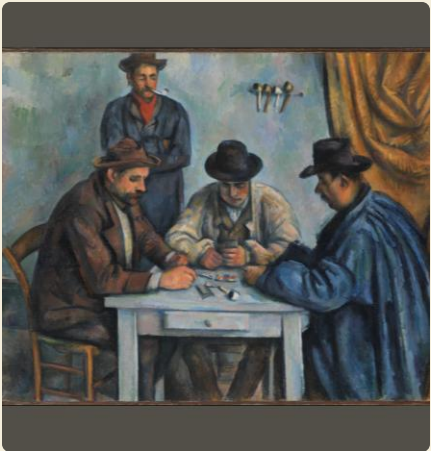
ARCHITECTURE / STRUCTURED

p026 View of a Village along a River



ABSTRACTION / SURREALISM

p039 Painting with Troika



HIGH-TEXTURE / BRUSHWORK

p043 The Card Players

EXPERIMENT DESIGN

CANONICAL

Four damage families plus zero-damage control across the collection

DAMAGE SIZE

Seven nested target levels on a focused, balanced subset

MASK ROBUSTNESS

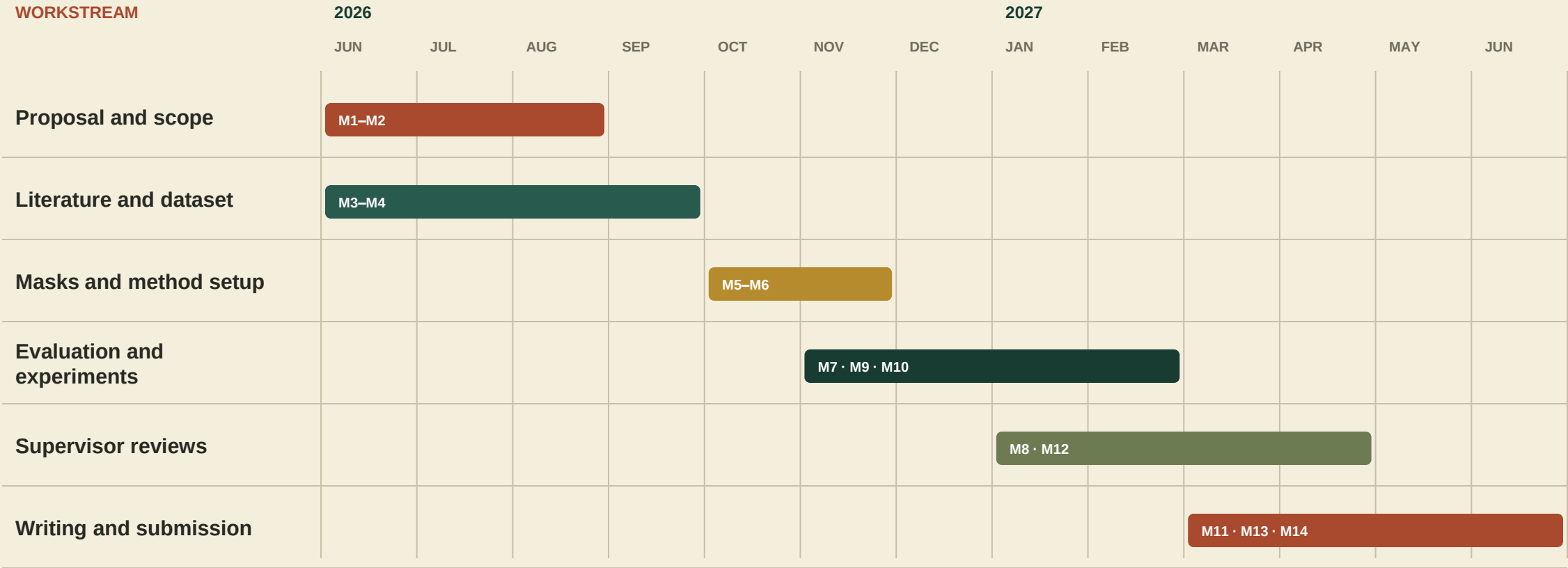
Matched placement and geometry changes on focused cases

DEGRADATION

Individual and combined procedural effects with task eligibility rules

Project Plan

June 2026 to June 2027, with review gates built in.



MILESTONE GATES

- M2 Proposal approved · 10 Sep 2026
- M6 Models selected and tested · 30 Nov 2026
- M8 Intermediate supervisor review · 15 Jan 2027
- M10 Uncertainty analysis and reporting · 28 Feb 2027
- M11 First thesis version · 31 Mar 2027
- M12 Final supervisor review · 15 Apr 2027
- M13 Final thesis handed in · 31 May 2027
- M14 Paper or extended abstract · 15 Jun 2027